

NLU course project: Language Modeling

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1. Introduction

I tackle word-level language modeling on the Penn TreeBank corpus, evaluated with perplexity (PPL). In Part A I train a GPT-2-style decoder-only Transformer [1] *from scratch*, improving it step by step through learning-rate and architecture search, dropout and a weight-tying ablation, lowering the test PPL from 35.68 to 33.07. In Part B I fine-tune the *pre-trained* GPT-2 with a hand-implemented LoRA, training only low-rank adapters on the Q, K, V projections while the 124M backbone stays frozen, reaching test PPL 19.14. Both parts satisfy $\text{PPL} < 250$, and Part B clearly benefits from starting from a pre-trained model.

2. Implementation details

Sentences are tokenized with the GPT-2 BPE tokenizer and padded on the right (padding ignored in the loss); PPL is the exponential of the per-token cross-entropy. Every experiment follows a greedy, one-change-at-a-time search, selected on *dev* PPL, with AdamW, gradient clipping and early stopping.

Part A. Search order: learning rate (on the smallest model), then the architecture one hyper-parameter at a time (`d_model`, heads, layers, FFN size), then dropout, then a weight-tying ablation (`lm_head` tied to the token embedding) [2]. Tying did *not* help here: the tied model needed 20 epochs to converge against 7 for the untied one, and reached a worse PPL despite $\sim 19\text{M}$ fewer parameters, so I keep the untied model as final. The fact that the tied model needs almost three times as many epochs to converge, while still reaching a worse PPL, suggests that the constraint makes optimization harder (conflicting gradients between the embedding and output-projection roles) more than the parameter reduction helps generalization – an effect likely tied to the combination of a pre-norm Transformer architecture and a small training corpus, unlike the LSTM setting of the original weight-tying study.

Part B. LoRA [3] is implemented manually: each attention layer injects low-rank matrices A, B into Q, K, V , computing $\Delta Wx = (\alpha/r)BAx$. $A \sim \mathcal{N}(0, 1)$, B is zero-initialized, so training starts exactly from the pre-trained model; only the adapters train (0.7% of parameters at rank 16). HuggingFace’s internal label shifting is used, so labels are not shifted manually as in Part A. Search order: learning rate (rank 4, $\alpha=8$ fixed), then rank $r \in \{4, 8, 16\}$ ($\alpha=2r$), then α at the best rank.

3. Results

Models are selected on dev PPL and scored on test PPL. Table 1 and Figure 1 trace Part A: every step except weight tying improves test PPL. Within the architecture step, doubling `n_heads` (4 $\rightarrow$ 8) helped at no parameter cost, while halving `num_layers` (4 $\rightarrow$ 2) clearly hurt (test PPL 37.57, not shown). Table 2 reports Part B: PPL decreases monotonically with rank, and a smaller-than-default scaling ($\alpha=8$ at rank 16) gives the best result, test PPL 19.14 – a 42% relative improvement over the best from-scratch model while training only 0.88M of 125M

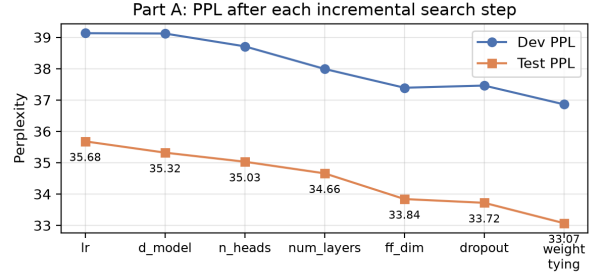


Figure 1: Part A: dev and test PPL after each incremental search step.

Table 1: Part A — best model per step. Final: $d=384$, 8 heads, 6 layers, FFN 2048, dropout 0.1, no weight tying.

Step	Config	Param	Dev	Test
0 – lr	$5 \cdot 10^{-4}$	29.2M	39.13	35.68
1.1 – d_model	384	44.6M	39.12	35.32
1.2 – heads	8	44.6M	38.71	35.03
1.3 – layers	6	47.3M	37.99	34.66
1.4 – FFN	2048	52.1M	37.39	33.84
2 – dropout	0.1	52.1M	37.46	33.72
3 – weight tying	not tied	52.1M	36.86	33.07

parameters.

4. References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [2] O. Press and L. Wolf, “Using the output embedding to improve language models,” in *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2017.
- [3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations (ICLR)*, 2022.

Table 2: *Part B — LoRA fine-tuning ($lr=5\cdot 10^{-4}$). 124.4M backbone frozen throughout.*

Rank r	α	Train.	Dev	Test
4	8	0.22M	23.27	21.02
8	16	0.44M	21.99	19.91
16	32	0.88M	21.20	19.35
16	8	0.88M	21.03	19.14